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Solar Receiver With a Porous Absorber

Abstract

A volumetric solar receiver is disclosed having a porous element with a pseudo-periodic lattice structure, wherein said porous element is arranged in the solar receiver to be heated by solar radiation and to transfer heat to a flowing medium. The porous element has a continuous functional gradient with reference to its structure in at least one direction pointing away from an outer region of the porous element arranged to be exposed to solar radiation. The gradient is such that the cell size of the porous element continuously decreases along at least one direction, from the outer region to an inner region of the element, without a discontinuity in the lattice pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is the United States national phase of International Patent Application No. PCT/EP2023/060678 filed Apr. 24, 2023, and claims priority to European Patent Application No. 22169935.8 filed Apr. 26, 2022, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND OF INVENTION

Field of the Invention

[0002] The invention is in the field of solar receivers, particularly for use in concentrated solar applications.

Description of Related Art

[0003] In a concentrated solar application, the solar radiation is concentrated and collected with the goal of electrical energy generation or use as thermal energy for industrial processes or for chemical reactions. Typical concentrated solar systems include solar towers wherein a number of heliostats are arranged to direct the solar radiation to a common receiver (usually installed on a tower) and parabolic systems wherein the solar radiation is concentrated in a focal point or line.

[0004] In a concentrated solar system, the concentrated solar radiation is absorbed by a receiver. Receivers can reach a very high temperature of many hundreds of degrees or even above 1000° C. The absorbed heat is transferred at such high temperature to a suitable medium, which is typically a liquid or a gas. The heated medium can be used as a heat source for example for an industrial process, for the production of electrical energy or for chemical reactions. The heated medium can be used as such or to transfer heat to another medium. For example, the heated medium can be used to produce steam for a steam turbine or, in some applications, the heated medium is air and the so obtained hot air can be directly expanded in a gas turbine. In other applications, the heat absorbed on the porous body of the solar receiver can be used for chemical reactions, for example for synfuel production.

[0005] A concentrated solar system needs a suitable device called solar receiver, which is arranged to absorb the concentrated solar radiation and be heated by said radiation, and to transfer heat to a flowing medium. In some concentrated solar systems, the solar receiver is made of a volumetric receiver, also called porous absorber or volumetric solar absorber, that is a porous material able to absorb the incident solar radiation and transfer the heat to a medium. The distinctive feature of a volumetric receiver is that the concentrated solar radiation is absorbed in the volume of the absorber, as the porous structure allows for penetration of the radiation. Desirable features of the absorber include, among others, open porosity, resistance to high temperature, sufficient mechanical strength, good absorption of the incident solar radiation and consequently a low power loss caused by reflection and a high efficiency in transferring the absorbed heat to a flowing medium.

[0006] In the current state-of-the-art, porous structures are regarded as best candidate for such volumetric receivers. Porous structures, also called cellular structures or lattice structures, are known and used for different industrial applications such as thermal insulation, structural applications, catalyst supports, heat exchangers and thermal energy storage. For application as volumetric receivers, porous structures having an engineered morphology appear particularly promising. The term engineered morphology identifies in particular the distribution of pores in the space, also named pore architecture or internal architecture. The thermal, optical and fluid-dynamic properties of such volumetric receivers are highly influenced by the morphological structure (including for example the pore distribution within the space) and, accordingly, there is an ongoing effort to identify the best structure for solar applications of interest.

[0007] A generally accepted classification of lattice structures is based on morphological

characteristics. A first distinction is made between stochastic (random) and non-stochastic lattices. A stochastic lattice structure consists of irregular and non-periodic cells resulting in a network of interconnected struts, surfaces, and/or plates wherein it is not possible to identify distinct features that are repeated in space. The cells of a stochastic lattice are randomly distributed in space and have different topological structures and sizes. An example of stochastic lattice is a foam. Non-stochastic lattices are made up of unit cells with distinct, repeatable features. Non-stochastic lattice structures can be further classified based on organization and periodicity. Organization-driven classification refers to the cellular organization of the lattice and includes strut-based, surface-based, plate-based, and honeycomb lattices. Periodicity-driven classification categorizes non-stochastic lattices into periodic and pseudo-periodic lattices. The periodicity of a lattice refers to the type and size of the unit cells throughout the structure. Periodic lattice structures are composed by a periodic repeated arrangement of unit cells with a certain shape, topology, and size in three-dimensional space. Pseudo-periodic lattice structures are composed of unit cells with the same topology but with different sizes and shapes. Examples of pseudo-periodic lattice structures are conformal and warped lattice structures. Periodic and pseudo-periodic lattices can be further categorized into homogeneous or heterogeneous lattices. Homogeneity refers to the thickness of the unit-cell elements such as struts and walls, therefore homogeneous lattice structures have a uniform strut and/or wall thickness and heterogeneous lattice structures have gradients in the strut and/or wall thickness.

[0008] Pseudo-periodic lattice structures and heterogeneous periodic lattices structures are part of a class of materials called functionally graded materials (FGM). Generally, FGM include all materials that present a variation in composition and/or structure over volume, resulting in corresponding changes in the properties of the material. The properties may change either continuously or in a discontinuous manner. With regards to the morphological characteristics of a lattice structure, a discontinuous FGM, also called discrete or layered FGM, is a lattice structure in which the structure changes in a stepwise mode, usually with the presence of an interface. A stepwise, or discontinuous change, is present when zones with different properties can be clearly identified. In contrast, a continuous FGM is characterized by a continuous change in structure with no clear separate zones and interfaces.

[0009] Relevant parameters of a lattice structure include, among others, the cell topology, cell morphology, the cell size, and the thickness of the cell elements such as struts and walls. There are many different cell types, including strut-based cells like kelvin, diamond, cubic, icosahedron, surface-based cells like gyroid, primitive, Schwarz diamond, honeycomb cells, among many others. The cell size is commonly defined by the values for width, length and height.

[0010] For several applications, the lattice structure properties of porosity and specific surface area are important. The porosity is defined as the ratio of void volume to the total volume of the structure. The specific surface area is the ratio of the total surface area to the total volume. For other applications, mechanical properties may be more important, such as compression and bending strength. In the case of volumetric receivers, as the structure must absorb the solar radiation to which it is directly exposed, also the optical properties of the structure play an important role. In particular, a high penetration of solar radiation into the volumetric receiver is regarded as beneficial for the heat exchange efficiency of the absorber.

[0011] Volumetric absorbers with a lattice structure can be made of several materials, including among others metal alloys, composites or ceramic and are obtainable, for example, by replica technique, extrusion, molding or additive manufacturing.

[0012] EP 3 359 886 discloses a volumetric solar receiver with hollow cells distributed over multiple levels. In a direction orthogonal to a face exposed to the solar radiation, the hollow cells of one level are smaller in size than hollow cells of the previous level, thus providing a stepped variation of the pore size of the receiver. It is suggested that said variation of the pore size allows a higher penetration of the solar radiation and better transfer of energy from the solar radiation to the

absorber. However, the stepped variation introduces a number of boundary regions (e.g. surfaces) within the structure of the receiver wherein the properties of the body change suddenly across the boundary region, i.e. show a discontinuity. These regions of discontinuity may represent structurally weak points exposed to the risk of local concentration of stress and failure of the assembly in particular in case of rapid temperature changes or in the presence of high temperature gradients within the structure. Furthermore, the regions of discontinuity may be affected by a poor heat exchange coefficient with the risk of local overheating.

[0013] Known lattice structures for volumetric receivers include structures based on the repetition of a unit cells, possibly with a variation of cell size within the structure, wherein the variation is obtained by assembling several lattice structures of each desired cell size to form a compound structure. Such a compound structure may provide a different cell size in different regions, however with the presence of discontinuous transitions from one structure to another (discontinuous FGM) and the drawbacks indicated above, i.e. possible formation of weak points and/or hot spots.

[0014] Still another known lattice structure is a stochastic lattice named Voronoi after the mathematical concept of Voronoi tessellation of space. This structure may also offer a variation of the cell size along a defined direction, however its intrinsic random arrangement of cells may create regions of hot spots or accumulation of stress in a substantially unpredictable manner.

[0015] Hence there is still the need to provide an optimized structure for volumetric receivers for concentrated solar systems.

SUMMARY OF THE INVENTION

[0016] The invention aims to provide a novel structure of volumetric solar receiver to increase the capture of concentrated solar radiation and to increase the heat transfer efficiency to a fluid medium such as gas or liquid crossing the absorber. Particularly, the invention aims to avoid the above-mentioned drawbacks of discontinuous features in the structure of the porous absorber.

[0017] The above aims are reached with a solar receiver according to claim 1. Certain preferred features are stated in the dependent claims.

[0018] The invention provides an absorber including at least one region having a lattice structure, wherein said lattice structure is made of a repetition in space of basic elements, the basic elements are connected to each other to define units of the structure, and the number of basic elements per unit is constant throughout said lattice structure. The units may be for example cells of a given morphology. Said units form orders next to each other according to a penetration depth of the incident solar radiation in the structure. Said depth in the structure may be regarded as a distance from a surface exposed to the solar radiation. The units are more densely arranged according to the depth in the structure and any element in common between adjacent units is shared entirely and not only partially.

[0019] Accordingly, for each pair of said orders comprising a first order and a second order next to each other, the second order being deeper than the first order in the structure, the following conditions are satisfied: [0020] in the second order, that is deeper than the first order, the units are more densely arranged than in the first order, so that the second order has a number of units per length or number of units per volume which is greater than the first order, and [0021] each unit of the first order has one or more basic element(s) which is shared with a unit of the second order wherein the one or more shared basic element(s) are shared along their entire extension between said units.

[0022] The above two conditions identify a structure which can be named pseudo-periodic. The structure has a continuity in the lattice pattern with no discontinuous change in morphology; the basic elements are connected to each other through the structure with continuity. The units of the structure can be unitary cells. According to the second condition above, any basic element which is shared between two units is shared entirely, to avoid that two basic elements intersect or connect at an intermediate location. This feature provides a continuity in the structure and avoids singularities which may become weak points of the structure.

DESCRIPTION OF THE INVENTION

[0023] A solar receiver according to the invention has at least one region with a peculiar lattice structure. The lattice structure of the invention is based on a repetition of basic elements, wherein the size of unitary cells delimited by said basic elements vary throughout the structure.

Accordingly, the structure can be regarded as a pseudo-periodic structure.

[0024] In certain embodiments, the basic elements are connected to each other to define unitary cells of the structure, wherein the number of basic elements and the number of connections and/or intersections between such basic elements, per each individual cell, is constant throughout the structure. Each individual cell is connected to at least one neighboring cell in at least one point. Additionally, there is at least one direction according to which the spatial arrangement of the basic elements and/or a size of the basic elements is modified to obtain a deformation of the unitary cells so that the individual volume of the cells decreases continuously along said direction.

[0025] Said direction may be regarded as a gradient of the size of the cells, as the cells become smaller along said direction. Said gradient direction points away from an outer region of the absorber, which is a region arranged to be exposed to the solar radiation when the receiver is in operation. Accordingly, the size of the cells decreases towards the inner of the absorber, i.e. the cells are larger in outer regions more exposed to the solar radiation and smaller in inner regions which receive less radiation.

[0026] The cells are delimited by a repetition in space of said basic elements, which may be regarded as elementary building blocks of the structure. The structure can be termed pseudo-periodic because the general topology of the cells (i.e. number of basic elements and related connections) per cell is invariant, although the shape and size of the cells may vary. Particularly the cells may be stretched, tapered or deformed throughout the structure.

[0027] The cells may be deformed in the above-mentioned gradient direction, whereas the continuity of the structure is preserved by the continuous distribution in space of the basic elements. The individual cells are connected to each other with a constant number of connections, and the location of the intercellular connection of two neighboring cells is the same, thus creating a continuous pattern without discontinuity.

[0028] In other embodiments said gradient is understood in terms of distance from a surface exposed to the incident solar radiation, or depth of the structure. In certain embodiments, the structure may be symmetrical around an axis and may be easier to represent with polar coordinates, such as cylindrical coordinates. For example, the distance from the receiving surface may be a radius in a cylindrical system.

[0029] A remarkable feature of the above defined pseudo-periodic lattice structure is the absence of regions of transition from one cell size to another (continuous FGM). Hence the invention avoids the drawbacks of structures resulting from assembly of parts, as explained above (discontinuous FGM). On the other hand, the structure maintains a higher degree of continuity compared to stochastic structures. Accordingly, the invention avoids the inherent drawbacks of randomly arranged foam-like or stochastic structures.

[0030] The applicant has found that the provision of such cell size gradient and the continuity of the lattice structure in combination with larger cells in the proximity of the outer region exposed to the solar radiation and smaller cells in the internal regions of the absorber improve the penetration of the solar radiation into the cellular structure, reduce back radiation, and improve mechanical resistance of the assembly. The fluid dynamic behavior of the system is also improved reaching a more suitable behavior for the intended application. The heat transfer efficiency between the porous element and the heat exchange medium that crosses the assembly is finally increased.

[0031] The solar receiver of the present invention can be used for various applications of concentrated solar technology, where a flowing medium is heated at high temperature (possibly greater than 1000° C.) and used for example for the production of electrical energy or as a heat source for a process, for example in heat demanding industrial processes or chemical reactors that

require heat to carry out endothermic reactions.

[0032] The flowing medium is preferably a gas or a liquid and more preferably is air. A stream of air can enter the porous absorber and heated to a suitable high temperature. The porous absorber absorbs heat from the incident solar radiation and transfers heat to the medium.

[0033] Said basic elements of the structure may include substantially mono-dimensional like linear elements or substantially bi-dimensional. Examples of substantially mono-dimensional elements are ligaments or struts. Examples of substantially bi-dimensional elements are surface-based structures. The connections between basic elements may include nodes where two or more basic elements converge and/or intersection lines between surfaces.

[0034] Preferred embodiments of the invention include a porous absorber with a strut-based pseudo-periodic structure or with a truss-based pseudo-periodic structure or with a surface-based pseudo-periodic structure.

[0035] An interesting embodiment of the invention is a strut-based structure wherein the basic elements are linear elements connected at their ends, to form nodes. The nodes delimit unitary cells of a given morphology, such as cubic cells or cells of a more elaborate shape. The cells are stretched or tapered so that their size decrease with the depth in the structure (penetration depth of the incident radiation); adjacent cells may share one or more struts and shared struts are entirely in common between adjacent units, so that all struts connect at their end points.

[0036] In a strut-based structure, each node constitutes an end point of a linear element and a vertex of a cell. Accordingly, the structure has no node which is at an end point of a basic linear element but no at a vertex of a cell, as such node would constitute a discontinuity in the structure.

[0037] Particularly preferably, each of said unitary cell is one of the following: a cubic cell, a parallelepiped cell, a diamond cell, an octahedron cell or a kelvin cell, an octahedron, a truncated octahedron, a gyroid, a triply periodic minimal surface-based cell. The above examples represent non-limiting examples.

[0038] The modification of the basic elements to decrease the size of cells may include any of: a uniform scaling, a non-uniform scaling, a deformation, a rotation or combinations thereof.

[0039] Preferably, each of said cells has linear dimensions comprised between 1 mm and 30 mm.

[0040] The unit cells may define the pore morphology and size, e.g. the pores are formed by voids between the structural elements, such as linear struts, that define the boundaries of the cells. The morphological characteristics of the lattice structure vary in a continuous way by varying spatial arrangement and size of the basic elements of the unit cell but keeping the same unit cell topology, as opposed to discontinuous or step-wise gradient, in which sections of a lattice structure with equal cell morphology but different cell sizes are assembled creating a discontinuity in the lattice pattern.

[0041] The continuous gradient may be achieved by realizing the porous absorber with a structure wherein unit cells are deformed, for example tapered, in at least one direction. For instance, using a reference cartesian coordinate system comprising an X, Y and Z-axis wherein the Z-axis is perpendicular to said outer region of the porous element, the unit cells may be deformed along one of said axis. Preferably, the unit cells are deformed along X- and Y-axis so to obtain a continuous cell size gradient along the Z-axis. More preferably, the unit cells are non-uniformly scaled along the X-, Y- and Z-axis.

[0042] The cell size may influence also the porosity of the absorber. Although the cell size and the porosity are distinct features, the variation of the cell size may, and in most case does, cause a corresponding variation of the porosity. According to the periodicity driven classification, homogenous structures are characterized by having a uniform strut and/or wall thickness. In such structures, a change in the cell size causes a change in the porosity, resulting in larger porosity in regions with larger cells and lower porosity in regions with smaller cells. Heterogeneous lattice structures have gradients in the strut and/or wall thickness, so it is theoretically possible to vary the cell size and keep the porosity constant by reducing the strut/wall thickness. The pore size of the

absorber is preferably between 50% and 95%.

[0043] Said outer region of the absorber may be for example a surface of the absorber exposed to the concentrated solar radiation. In some embodiments the gradient direction according to which the size of the unit decreases, is a direction perpendicular to the incident solar radiation.

[0044] The porous absorber can be made, among others, by molding, replica technique or with an additive manufacturing technique. The additive manufacturing shaping process may be performed directly from a CAD file. Particularly, the porous absorber may be made of any of: a metal, a metal alloy, a composite, a ceramic material, the ceramic material being preferably silicon carbide.

[0045] The porous element can be manufactured with various materials or alloys depending on the final application of the solar receivers. Preferably, for high-temperature applications the porous element is made of ceramic and more preferably is made with silicon carbide.

[0046] Other aspects of the invention are defined in the following paragraphs. [0047] i) A solar receiver comprising a porous absorber arranged to be heated by solar radiation and to transfer heat to a flowing medium, wherein the absorber includes at least one region having a pseudo-periodic lattice structure, said structure being made of basic elements such as struts for example, wherein the basic elements form unitary cells, the structure having at least one gradient direction, pointing away from an outer region of the absorber exposed to the solar radiation, wherein according to said gradient direction the arrangement in space of said basic elements and/or the size of said basic elements vary so to deform the unitary cells so that the individual volume of the cells decreases continuously along said direction while maintaining continuity in the lattice pattern with no discontinuous change in morphology. [0048] ii) A solar receiver comprising a porous absorber arranged to be heated by solar radiation and to transfer heat to a flowing medium, wherein the absorber includes at least one region having a pseudo-periodic lattice structure, said structure being made of basic elements arranged to form unitary cells, the structure having at least one gradient direction, pointing away from an outer region of the absorber exposed to the solar radiation, wherein according to said gradient direction the spatial density of the unitary cells increases, wherein each unitary cell shares one or more basic element with adjacent unitary cells, wherein any basic element, which is shared between adjacent unitary cells, is shared entirely between said cells. [0049] iii) A solar receiver comprising a porous absorber arranged to be heated by solar radiation and to transfer heat to a flowing medium, wherein the absorber includes at least one region having a pseudo-periodic lattice structure, said structure being made of basic elements arranged to form unitary cells, the structure having at least one gradient direction, pointing away from an outer region of the absorber exposed to the solar radiation, wherein according to said gradient direction the spatial density of the unitary cells increases, wherein the spatial density of the unitary cells is increased by a modification of the basic elements which includes one or more of: a uniform scaling, a non-uniform scaling, a deformation, a rotation or combinations thereof.

[0050] The features in the above aspects i), ii) and iii) may be combined into further embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0051] FIG. 1 shows a schematic representation of a solar receiver according to an embodiment of the invention.

[0052] FIG. 2a is a cross section view of a lattice structure with a stepwise change of the cell size through layers of an absorber of the prior art.

[0053] FIG. 2b is a cross section view of a lattice structure with a continuous pattern with cell size reduction in a preferred direction which is in accordance with an embodiment of the invention.

[0054] FIG. 2c is a 3D model of a discontinuous functionally graded material according to prior

art.

[0055] FIG. **2d** is a 3D model of a continuous functionally graded material in accordance with an embodiment of the invention.

[0056] FIG. **2e** is a magnified view of a unit of a lattice structure in accordance with an embodiment of the invention.

[0057] FIG. **3** illustrates the structure of a porous absorber according to an embodiment of the invention wherein the unit cells of the absorber are diamond cells.

[0058] FIG. **4** illustrates the structure of a porous absorber according to an embodiment of the invention wherein the unit cells of the absorber are octahedron cells.

[0059] FIG. **5** illustrates the structure of a porous absorber according to an embodiment of the invention wherein the unit cells of the absorber are Kelvin cells.

[0060] FIG. **6a** illustrates one exemplary deformed Kelvin cell of the structure of FIG. **5**.

[0061] FIG. **6b** illustrates one regular Kelvin cell that can be used in a periodic lattice structure or in a discontinuous functionally graded material.

[0062] FIG. **7** illustrates an example of a surface-bases periodic cellular structure of the so called gyroid cell type.

[0063] FIG. **8** is a sectional view of a solar receiver according to an embodiment.

[0064] FIG. **9** is a sectional view of a solar receiver according to another embodiment.

DETAILED DESCRIPTION

[0065] FIG. **1** illustrates a volumetric receiver **1** comprising a body **8** and a porous absorber **3** with a receiving surface **2** exposed to the incident solar radiation **7**.

[0066] The solar radiation **7** may be a concentrated solar radiation wherein concentration is performed with known techniques such as the use of parabolic dish concentrators or an array of heliostats.

[0067] The body **8** in the present embodiment is represented for simplicity with a rectangle box but depending on the specific solar receiver design the body **8** may have other shapes.

[0068] The porous absorber **3** is contained within the body **8** and is crossed by a flowing medium **4** that enters the receiver **1** via the receiving surface **2** and leaves the receiver via an outlet surface **5** as a hot medium **6**.

[0069] The receiving surface **2** is directly exposed to the incident radiation **7**, so that the porous absorber **3** can be heated by the radiation **7** and transfer heat to the medium **4**.

[0070] The porous absorber **3** is at least in part made with a pseudo-periodic lattice structure and has a continuous functional gradient with reference to its structure in the negative direction of the Z-axis. The size of the cells of the absorber **3**, accordingly, decreases along the above-mentioned direction from the receiving surface **2** towards the outlet surface **5** without a discontinuity in the lattice pattern.

[0071] For a better understanding of the continuous functional gradient of the invention, reference can be made to FIG. **2** wherein FIG. **2a** and FIG. **2c** illustrate a discontinuously graded structure and FIGS. **2b**, **2d** illustrate a continuously graded structure.

[0072] FIGS. **2a** and **2c** illustrate an example of cell size variation obtainable with an assembly of structures wherein each zone of the structure has a constant cell size, according to the prior art. The cell size is constant across each zone and decreases from one zone to the next thus resulting in a stepped variation across the porous absorber.

[0073] Steps between zones **Z1** and **Z2** are indicated in the strut-based structure of FIG. **2a** and FIG. **2c**. The steps are formed where a size of a cell **33** located in a zone **Z1** is not equal to the size of a cell **34** in the next zone **Z2**, and therefore a discontinuity in cells size is formed between zones of the structure. For example, FIG. **2a** illustrates that two cells **34** in the zone **Z2** are adjacent to a single cell **33** in zone **Z1** and, consequently, intersections **35** are formed at intermediate points of struts.

[0074] Hence the functionally graded material is said to be discontinuously graded because the

morphological characteristics are constant within a zone but vary suddenly when passing from one zone to another. An interface is clearly recognisable with a stepped variation which, as above discussed, may lead to the formation of weak points wherein the thermal and/or mechanical stress can accumulate potentially leading to the failure of the component and leads to sub-optimal heat exchange.

[0075] By comparison, the structure of FIGS. **2b** and **2d**, which is in accordance to an embodiment of the invention, has no such discontinuity and no intermediate intersection like the intersections **35** of the prior-art structure. These embodiments will be further described below.

[0076] The lattice structure of the present invention provides a continuous change in the morphological characteristics along a preferred direction according to an embodiment of the present invention. The continuity of the lattice pattern is preserved, and the size of the cells decreases within the structure without a locally concentrated variation at the interface between zones. Hence the functionally graded material is said to be continuously graded because the morphological characteristics are changing in a continuous way without separate zones and interfaces. The size of cells is greater in the proximity of the outer surface exposed to the solar radiation, and decreases within the structure of the absorber away from the surface exposed to the solar radiation such as the surface **2** in the figures.

[0077] FIGS. **3** to **7** illustrate examples of the structure of a porous absorber of a solar collector according to various embodiments of the invention. FIGS. **3** to **6** disclose strut-based structures, whereas FIG. **7** is an example of a surface-based structure.

[0078] FIG. **3** shows a lateral view of the structure of a porous absorber made of diamonds unit cells. In the figure, the unit cells are arranged to create a continuous lattice pattern with cell size gradient in the negative direction of the Z-axis. The vector **13** in FIG. **3** denotes the direction of the cell size gradient.

[0079] The diamond unit cells of FIG. **3** are made by linear elements or struts **T** which are interconnected at nodes **N**. Each cell shares a strut **T** with an adjacent next cell, for example the cells **C1** and **C2** share the strut **T1**. Each cell defines a pore **P** of the porous body of the absorber. In accordance with the gradient **13**, each cell is smaller than cells upstream and larger than cells downstream, e.g. the cell **C2** is smaller than the cell **C1**, and consequently the pores **P** become smaller according to the direction of the gradient **13**.

[0080] It can be appreciated in FIG. **3** that the continuity of the structure is preserved despite the variation of the cell size, because the cells are interconnected to each other by the struts **T**. The cell size varies in the upstream-downstream direction identified by the gradient **13**. The structure does not have macroscopic boundaries of transition between regions with different pore sizes.

[0081] FIG. **4** and FIG. **5** show, similarly to FIG. **3**, an embodiment with octahedron cells and an embodiment with Kelvin cells. Also in these embodiments, each cell shares at least one strut with an adjacent cell so that the structure has an overall continuity, and the cells and pore size become smaller in the upstream-downstream direction of the gradient **13**, which in the examples is still the negative-Z.

[0082] FIG. **6a** illustrates one exemplary deformed Kelvin cell of the structure of FIG. **5**. FIG. **6b** illustrates one regular (non-deformed) Kelvin cell used in a periodic lattice structure or in a discontinuous functionally graded material. Particularly, the cell of FIG. **6a** is stretched in the Z direction compared to the cell of FIG. **6b**.

[0083] FIG. **6** provides an example of a cell deformation in the gradient direction, suitable to obtain a continuous variation of the cell size within the structure of the absorber **3**.

[0084] FIG. **7a** provides a cross section view of a porous element made with a surface-based cellular structure comprising a plurality of gyroid unit cells. FIG. **7b** provides a 3D model of the same gyroid unit cell. As in the previous embodiment, the gyroid unit cells are arranged to create a cell size gradient in the negative direction of the Z-axis by preserving the lattice pattern, i.e. with no discontinuity.

[0085] The lattice structure **9** is hereafter disclosed in more detail with reference to FIGS. **2b**, **2d**, **2e**, **3-7**.

[0086] The lattice structure **9** is made of a repetition in space of basic elements **10**, wherein the basic elements **10** are connected to each other to define units **11** of the structure **9**, and the number of basic elements **10** per unit **11** is constant throughout the lattice structure **9**.

[0087] In the example of FIG. **2b**, each unit **11** is formed by four basic elements **10** in the form of struts. The basic elements **10** are connected at connection points N to define quadrangular shapes having upper basic elements **10a** and lower basic elements **10b** (of the basic elements **10**) disposed in parallel. The units **11** are arranged in a plurality of rows such as the rows **12a**, **12b**, wherein upper basic elements **10a** and lower basic elements **10b** of units **11** in a same row **12a** are aligned. Hence the orders of the units **11** are represented, in this embodiment, by rows such as the rows **12a**, **12b**.

[0088] In the example of FIG. **2d**, each unit **11** is formed by twelve basic elements **10**. A magnified view of one unit **11** is represented in FIG. **2e**. Four upper basic elements **10a** and four lower basic elements **10b** delimit, respectively, an upper quadrangular face **10A** and a lower quadrangular face **10B**. Said faces **10A**, **10B** are parallel to each other and connected by the remaining four basic elements **10c** of the unit **11**.

[0089] The units **11** are ordered in a plurality of rows **12a**, **12b**, wherein the upper quadrangular shapes **10A** and lower quadrangular shapes **10B** of units **11** in each row are coplanar. The plurality of rows are further ordered in columns such as columns **15a**, **15b** to form matrixes of units **11**.

[0090] Whereas a parallelism and coplanarity is defined in the patterns of units **11** of FIGS. **2b** and **2d**, patterns of basic elements **10** has not (necessarily) constrains of this type.

[0091] In FIG. **3**, for instance, each unit **11** still includes four basic elements **10** as struts, defining quadrangular shapes. However, basic elements **10** are not arranged in parallel. To the contrary, units **11** are in ordered such that upper vertexes Na and lower vertexes Nb of units **11** in each row **12b** are aligned along an upper line and a lower line of the order, and intermediate vertexes Nc are aligned along an intermediate line. Upper vertexes Na and lower vertexes Nb of units **11** in each row **12b** are aligned. Preferably, each one of the basic elements **10c** of one unit **11** has a radius of curvature Rc in common with other basic elements **10c** of units **11** not belonging to the same row **12b**.

[0092] In the embodiment of FIG. **4**, the lattice structure **9** of FIG. **2b** and FIG. **3** are overlapped so as the vertexes Na, Nb (of the lattice structure **9** of FIG. **3**) are aligned on upper basic elements **10a** and lower basic elements **10b** of the lattice structure of FIG. **2b**. The units **11** of the resulting lattice structure **9** of FIG. **4** includes therefore eight struts **10**.

[0093] FIG. **5** is an example wherein each unit **11** includes sixteen basic elements **10** (struts). Eight basic elements **10** are connected to define polygons POL, wherein, similarly to the embodiment of FIG. **2b**, upper basic elements **10a** and lower basic elements **10b** are parallel and the units **11** are ordered in a plurality of rows (or orders) **12a**, **12b**, wherein the upper basic elements **10a** and lower basic elements **10b** of units **11a** in each row **12a** are aligned. The remaining eight basic elements **10** define a quadrangular shape Q inside the polygon POL, which is connected to the sides **10** of the polygon POL by means of connecting struts **10d**. Said connecting struts **10d** extend, preferably, between a middle point of one side **10** of the polygon POL to a vertex Nc of the quadrangular shape Q.

[0094] In the description provided above with reference to FIGS. **2** to **5**, the orders of units have been disclosed as rows, for easy of explanation. In the example, units **11** are arranged to form orders **12a**, **12b** of units **11** next to each other. However in other embodiments of the invention the orders may represent different arrangements other than rows, such as layers in the structure.

[0095] The orders **12a**, **12b** next to each other are arranged according to a depth **13** of the structure **9**, wherein the depth is a distance from a surface **14** exposed to the solar radiation **7**.

[0096] According to FIG. **2d**, for instance, the units **11** are arranged in the orders **12a**, **12b** based on

the depth **13** from the surface **14**. The same may apply to other embodiments.

[0097] Each pair of orders **12a**, **12b**, comprising a first order **12a** and a second order **12b** next to each other, with the second order **12b** deeper than the first order **12a**, satisfies the following: [0098] a) in the second order **12b**, that is deeper than the first order **12a**, the units **11** are more densely arranged than in the first order **12a**, so that the second order **12b** has a number of units **11** per length or number of units **11** per volume which is greater than the first order **12a**, [0099] b) each unit **11a** of the first order **12a** has one or more basic element(s) **10'** which is shared with a unit **11b** of the second order **12b** wherein the one or more shared basic element(s) **10'** are shared along their entire extension between said units **11**.

[0100] Accordingly, the lattice structure **9** is continuous. Indeed, no steps is formed between units of an order and unit of a deeper (next order). The unit of an order ends exactly where the unit of the next order starts, sharing with it the basic elements, and therefore not involving further units in the direction of depth of the structure. At the same time, due to the greater density at the next order, a size of the units is decreased in the direction of depth.

[0101] This is achieved for instance with units having a shape which is tapered in the direction of depth. For instance, with reference to FIG. **6b**, a unit includes basic elements are arranged to delimit in the periphery of the unit a regular octagon which is not adapted to satisfy the condition of the present invention.

[0102] FIG. **6a**, instead, schematically represents a unit including basic elements delimiting in the periphery of the unit an octagon with a tapered shape, wherein a length of a first basic element (the top element in the drawing) forming the octagon is more than the length of an opposed second basic element (the bottom element) of the octagon. Therefore, the unit as represented in FIG. **6a** is suitable to be a unit of an order according to the present invention, since the second basic element (bottom element) of the unit may be shared with a unit of next order (not represented), in its entire length, becoming the top element of the unit of the next order, with no discontinuities between the units.

[0103] As to the density, considering for instance the embodiment of FIG. **2b**, given a reference length **L** in the lattice structure **9**, the order **12a** includes four units **11** within said length **L** and the order **12b**, which is deeper than the order **12a**, includes five units **11** in the same length **L**. Considering element sharing, the unit **11a** of the first order **12a** has one basic element **10'** which is shared with the unit **11b** of the second order **12b** along its entire extension between the units **12a**, **12b**. The shared element **10'** has a length smaller than a length of the parallel basic elements **10a** in the unit **11a** and greater than a length of the parallel basic element **10c** in the unit **11b**.

[0104] In embodiment of FIG. **2d**, given the length **L** per volume in the lattice structure **9**, the order **12a** includes six units **11**, whereas the order **12b**, which is deeper than order **12a**, includes eight units **11** (density). The unit **11a** of the first order **12a** has one basic element **10'** which is shared with the unit **11b** of the second order **12b** along its entire extension between the units **12a**, **12b** (element sharing). In this embodiment, a plurality of element **10'** are shared, in particular four elements. The four shared basic elements define a quadrangular shape. The quadrangular shape is shared among units **11a** and **11b**. The shared quadrangular shape has an area less than an area of the parallel quadrangular shape in the unit **11a** and more than an area of the parallel quadrangular shape in the unit **11b**. Therefore, element(s) sharing may include more than one basic element or more complex structure.

[0105] In the embodiments of FIG. **4**, it may be appreciated that the density (number of units per length) increases with the depth **13** of the structure **9**. Moreover, each unit **11a** in a row **12a** shares two basic elements **10'** with two adjacent units **11b**, **11c** belonging to a next, deeper row **12b**. The conditions a), b) mentioned before are satisfied in the example of FIG. **5** as well, being a superimposition of embodiments already disclosed.

[0106] As to the example of FIG. **6**, a basic element **10'** is at the bottom side of the polygon POL of the unit **11a** in the row **12a**, and corresponds to the top side of the polygon POL of the unit **11b** in

the row **12b**, next and deeper row **12a**. The term row here is used instead of the more generic term “order”, since sort of rows **12a**, **12b** may be easily identified in FIG. 6.

[0107] Preferred featured of the lattice structure **9** are disclosed below.

[0108] Each unit **11** includes a plurality of connections N between basic elements **10** of the unit **11**, and the number of connections N inside each unit **11** is the same. In other words, the basic elements **10** have a same “intra-connection” (the connection inside the unit).

[0109] In the embodiment of FIG. **2b** four connections N are provided; in the embodiment of FIG. **2d**, eight connections are provided; in the embodiment of FIGS. **3** to **5**, respectively, four, eight and sixteen connections are provided. The same number of basic elements and connections per units allows to keep a same morphology for units at different depth.

[0110] In a preferred embodiment, each first unit **11** in the first order **12a** and each second unit **11** adjacent to said first unit **11** in the same order **12a** are connected to each other. In the embodiment of FIG. **2b**, connection of two units at same order is made by means of a basic element **10** shared between the units **11** in said same order, in particular by sharing the entire length of the basic elements **10**. The same apply to the embodiment of FIG. **2d**, wherein however the connection of two units **11** at same order is made by means of a plurality of basic elements **10**, forming a quadrangular shape which, is therefore also a shared shape between the two adjacent units **11** at the same order **12a**.

[0111] All the basic elements **10** of a unit **11** may be also shared with basic elements **10** of other units **11**, at same order or at different orders. This may be immediately appreciated from FIGS. **1** to **3**, for instance. Sharing all the elements increases heat transmission uniformity in the direction of depth of the structure and along the extension of each order **12a**, **12b**.

[0112] When not all the basic elements **10** of one unit **11** are shared with other units, such as for instance in the example of FIGS. **4**, **5**, preferably, the number of basic elements **10** shared with the other units **11** in the same or in a different order(s) is the same. Moreover, the number of connections N of a unit **11** with other units **11** in the same or in different order(s) is the same. This feature is also said “inter connection” (connection outside the unit). For instance, each unit **11** in the example of FIG. **4** shares four basic elements **10** with adjacent cells (taken from units at the same and/or different orders) and eight connections; each unit **11** in the example of FIG. **5** shares eight basic elements **10** with adjacent units **11** (taken from the same and/or different orders) and sixteen connections.

[0113] The examples above are disclosed mainly with reference to basic elements **10** in the form or struct or rectilinear elements.

[0114] The basic elements **10** within the meaning of the present invention may also include surface elements. This is for instance the case of the non-limiting example of FIGS. **7a** and **7b**. As schematically represented in the cross section of FIG. **7a**, orders **12a** and **12b** are defined wherein, in each order **12a** or **12b**, a plurality of units **11** are arranged. The density of units **11b** in the next order **12b**, i.e. in the order **12b** at a greater depth than order **12a**, is greater than the density of units **11a** in order **12a**. The units **11a**, **11b** are formed by surface elements **10**, for instance curved. The surface elements **10** are connected in a same unit **11a** by means of connection lines N or border regions N. The number of connection lines N among surface elements **10** per unit **11** is preferably the same. The unit **11** of an order **12a** is connected to the unit **11b** of the next order **12b** by means of at least one surface element **10**. The surface element **10** is shared among the units **11a** and **11b** on its overall surface extension.

[0115] As to the arrangement in the solar receiver **1**, the lattice structure **9** may be located in one single region of the absorber **3** (FIG. **1**) or in a plurality of regions (FIGS. **8** and **9**).

[0116] FIGS. **8** and **9** are examples of arrangement of the lattice structure **9** in the absorber of the solar receiver **1**, wherein the flowing medium **4** enters through an inlet and exit through an outlet as a hot medium **6**, defined in the body **8** of the solar receiver **1**.

[0117] In the example given with reference to the cross section of FIG. **8**, the lattice structure **9** is a

cylindrical structure arranged in a cavity **40**. The solar radiation enters through an aperture **41** of said cavity **40**. The structure **9** includes a base **15** distanced from said aperture **41**, and an annulus **16**. The structure **9** is contained in a body **8** with thermal insulation **42**.

[0118] In the base **15** the orders of the units **11** may be vertically arranged such as orders **15a**, **15b** and **15c**. For example, units **11** of an order **15a** are closer to a top surface **15''** of the base portion **15** (closer to the inlet), and units **11** of the orders **15b** or **15c** of higher density are closer to the bottom part **15'** of the base **15** (closer to the outlet). The depth of an order can be regarded as the distance from the surface **15''** exposed to the solar radiation.

[0119] In the side annulus **16** the orders of units **11** may be radially arranged such as orders **16a**, **16b** and **16c**. The orders may be regarded as crowns or cylindrical layers. The depth of an order in this case is the distance from an inner surface **16''** of the annulus **16**, which is exposed to the solar radiation entering the cavity **40**. Thus, units **11** of a next order **16b** of higher density are closer to an outer surface **16'** of the annulus **16**, and units **11** of the order **16a** of lower density are closer to the inner part **16''** of the annulus **16**.

[0120] Another arrangement of orders is disclosed for instance in FIG. **9**. Here, the structure **9** is contained in a body **8** with thermal insulation layer **52** and the solar radiation enters from above via a window **50**. The structure **9** includes a disc-like portion **25** and a side portion **26**. The side portion **26** includes segments having a different inclination with respect to the portion **25**. The thermal fluid travels through channels **51** made in the body **52**. The orders of the units in the structure may be in accordance with a radial distance from the window **50** or in accordance with the thickness of the structure **9** in both portions **25** and **26**.

[0121] In general, the orders of units in the absorber structure **9** may be adapted based on the structure of the body **8** and/or based on path of solar irradiation, to increase the heating of the flowing medium downstream the deepest order.

[0122] Based on the structure of the body **8**, inlet and the outlet of the flowing medium may be arranged on a same side of the body. It is not necessary to arrange the inlet of the flowing medium at the same side where the absorber is exposed to the solar radiation.

[0123] Although, the examples above have been given with reference to crowns as possible orders of units, further curvilinear or rectilinear orders of units may be defined and are covered by the scope of protection of the claims.

Claims

1. A volumetric solar receiver comprising a porous absorber to be heated by solar radiation and to transfer heat to a flowing medium fed to said volumetric solar receiver, wherein: the absorber is arranged in the volumetric solar receiver to be exposed to solar radiation, the absorber includes at least one region having a lattice structure, wherein said lattice structure is made of a repetition in space of basic elements, the basic elements are connected to each other to define units of the structure, and the number of basic elements per unit is constant throughout said lattice structure; in the lattice structure, said units are arranged to form orders of units, wherein the orders of units are arranged next to each other according to a depth in the structure, wherein said depth in the structure is a distance from a surface exposed to the solar radiation; each pair of said orders, comprising a first order and a second order next to each other, the second order being deeper than the first order in the structure, satisfies the following conditions: a) in the second order, that is deeper than the first order, the units are more densely arranged than in the first order, so that the second order has a number of units per length or number of units per volume which is greater than the first order, b) each unit of the first order has one or more basic element(s) which is shared with a unit of the second order wherein any basic element(s) shared between a unit of the first order and a unit of the second order is shared along its entire extension between said units.

2. The solar receiver of claim 1, wherein each unit comprises a plurality of connections between

basic elements of the unit, wherein the number of connections between basic elements inside each unit is the same.

3. The solar receiver of claim 1, wherein each unit in one of said orders is connected to at least another unit in the same one of said orders.

4. The solar receiver of claim 3, wherein said each unit and at least another unit are connected by means of at least one basic element.

5. The solar receiver of claim 4, wherein said at least one basic element is shared along its entire extension between said each unit and at least another unit in said same one order.

6. The solar receiver of claim 3, wherein all the shared basic elements of said each unit are shared along their entire extension between other units in said same one order or in another order.

7. The solar receiver of claim 3, wherein each unit is connected to another unit in said same one order or in another order by means of at least one connection.

8. The solar receiver of claim 7, wherein the number of connections between each unit and other units in said same one order or in another order is the same.

9. The solar receiver of claim 7, wherein said connection between each unit and other units in said same one order or in another order is at an end or a vertex of a basic element.

10. The solar receiver of claim 1, wherein each unit is tapered in a direction of said depth.

11. The solar receiver of claim 10, wherein said basic elements are surfaces, and wherein said connections are border regions between said surfaces.

12. The solar receiver of claim 1, comprising different type of orders, said type comprising at least one among rows, matrixes or crowns.

13. The solar receiver of any of the previous claims, wherein all the basic elements of each unit of an intermediate order are shared along their entire extension with another unit of the same order or of an adjacent order.

14. The solar receiver according to claim 1 wherein: the basic elements are connected to each other to define unitary cells of the structure, wherein the number of basic elements and the number of connections and/or intersections between such basic elements, per each individual cell, is constant throughout the structure.

15. The solar receiver according to claim 14 wherein each individual cell is connected to at least one neighboring cell in at least one point; there is at least one direction according to which the spatial arrangement of the basic elements and/or a size of the basic elements is modified to obtain a deformation of the unitary cells so that the individual volume of the cells decreases continuously along said direction, said direction points away from an outer region of the absorber, which is a region arranged to be exposed to the solar radiation when the receiver is in operation.

16. The solar receiver of claim 15 wherein the cells are stretched, tapered or deformed throughout the structure.

17. The solar receiver of claim 1 wherein said basic elements of the structure are substantially mono-dimensional like linear elements or substantially bi-dimensional.

18. The solar receiver of claim 1 wherein the basic elements are linear elements connected at their ends, to form nodes; the nodes delimit unitary cells of a given morphology, such as cubic cells or cells of a more elaborate shape; the cells are stretched or tapered so that their size decrease with the depth in the structure; adjacent cells share one or more struts and shared struts are entirely in common between adjacent units, so that all struts connect at their end points.

19. The solar receiver of claim 18 wherein each node constitutes an end point of a linear element and a vertex of a cell.

20. The solar receiver of claim 1 wherein each unitary cell is one of the following: a cubic cell, a parallelepiped cell, a diamond cell, an octahedron cell or a kelvin cell, an octahedron, a truncated octahedron, a gyroid, a triply periodic minimal surface-based cell.

21. The solar receiver of claim 1 wherein a modification of the basic elements to decrease the size and increase the spatial density of the unitary cells comprises any of: a uniform scaling, a non-

uniform scaling, a deformation, a rotation or combinations thereof.

22. The solar receiver of claim 1 wherein the unitary cells are non-uniformly scaled along three axes X, Y and Z defining an orthogonal reference system.
